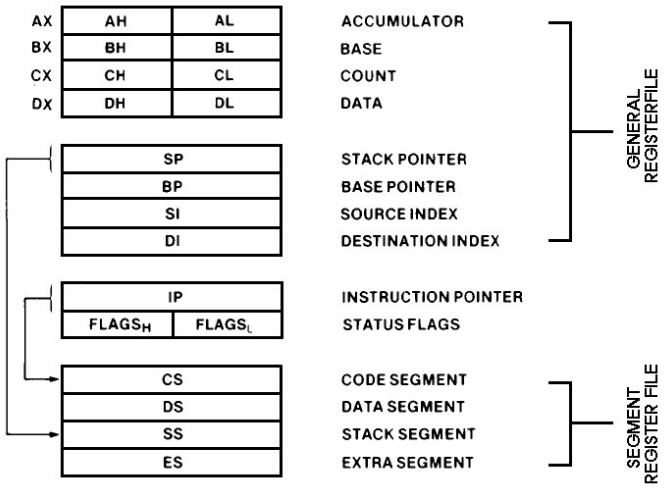
4. 8086 Instruction Set Summary.

8086
REGISTER MODEL



Instructions which reference the flag register file as a 16-bit object use the symbol FLAGS to represent the file:

FLAGS = X:X:X:X:(OF):(DF):(IF):(TF):(SF):(ZF):X:(AF):X:(PF):X:(CF)

x = don't care

AF : AUXILIARY CARRY - BCD
CF : CARRY FLAG
PF : PARITY FLAG
SF : SIGN FLAG
ZF : ZERO FLAG

DF : DIRECTION FLAG [STRINGS]
IF : INTERRUPT ENABLE FLAG
OF : OVERFLOW FLAG [DF & SF]
TF : TRAP - SINGLE STEP FLAG

OPERAND SUMMARY

"reg" field bit assignments :

16-Bit (w = 1)	8-Bit (w = 0)	Segment
000 AX	000 AL	00 ES
001 CX	001 CL	01 CS
010 DX	010 DL	10 SS
011 BX	011 BL	11 DS
100 SP	100 AH	
101 BP	101 CH	
110 SI	110 DH	
111 DI	111 BH	

SECOND INSTRUCTION BYTE SUMMARY

mod	xxx	r/m

mod	Displacement
00	DISP : 0". disp-low and disp-high are absent
01	DISP : disp-low sign-extended to 16-bits. disp-high is absent
10	DISP = disp-high ; disp-low
11	r/w is treated as a "reg" field

r/m	Operand Address
000	(BX) + (SI) + DISP
011	(BX) + (DI) + DISP
010	(BP) + (SI) + DISP
011	(BP) + (DI) + DISP
100	(SI) + DISP
101	(DI) + DISP
110	(BP) + DISP
111	(BX) + DISP

DISP follows 2nd byte of instruction (before data if required)

* except if mod = 00 and r/m = 110 then EA = disp-high; disp-row

Operand address (EA) Timing (clocks):

Add 4 clocks for word operands at ODD ADDRESSES

immed offset = 6

Base(BX, BP, SI, DI) = 5

Base + DISP = 9

Base + Index (BP + DI, BX + SI) = 7

Base + Index (BP + SI, BX + DI) = 8

Base + Index (BP + DI, BX + SI) + DISP = 11

Base + Index (BP + SI, BX + DI) + DISP = 12

DATA TRANSFER				
MOV = Move:	**76543210**	**76543210**	**76543210**	**76543210**
Register/Memory to/from Register	100010dw	mod reg r/m		
Immediate to Register/Memory	1100011w	mod000r/m	data	data if w = 1
Immediate to Register	1011wreg	data	data if w = 1	
Memory to Accumulator	1010000w	addr-low	addr-high	
Accumulator to Memory	1010001w	addr-low	addr-high	
Register/Memory to Segment Register	10001110	mod 0 reg r/m		
Segment Register to Register/Memory	10001100	mod 0 reg r/m		
PUSH = Push:				
Register/Memory	11111111	mod110r/m		
Register	01010reg			
Segment Register	000reg110			
POP = Pop:				
Register/Memory	10001111	mod000r/m		
Register	01011reg			
Segment Register	000reg111			
XCHG = Exchange:				
Register/Memory with Register	1000011w	mod reg r/m		
Register with Accumulator	10010reg			
IN = Input from:				
Fixed Port	1110010w	port		
Variable Port	1110110w			
OUT = Output to:				
Fixed Port	1110011w	port		
Variable Port	1110111w			
XLAT = Translate Byte to AL	11010111			
LEA = Load EA to Register	10001101	mod reg r/m		
LDS = Load Pointer to DS	11000101	mod reg r/m		
LES = Load Pointer to ES	11000100	mod reg r/m		
LAHF = Load AH with Flags	10011111			
SAHF = Store AH into Flags	10011110			
PUSHF = Push Flags	10011100			
POPF = Pop Flags	10011101			

ARITHMETIC	**76543210**	**76543210**	**76543210**	**76543210**
ADD = Add:				
Reg./Memory with Register to Either	000000dw	mod reg r/m		
Immediate to Register/Memory	100000sw	mod000r/m	data	data if s: w = 01
Immediate to Accumulator	0000010w	data	data if w = 1	
ADC = Add with Carry:				
Reg./Memory with Register to Either	000100dw	mod reg r/m		
Immediate to Register/Memory	100000sw	mod010r/m	data	data if s: w = 01
Immediate to Accumulator	0001010w	data	data if w = 1	
INC = Increment:				
Register/Memory	1111111w	mod000r/m		
Register	01000reg			
AAA = ASCII Adjust for Add	00110111			
BAA = Decimal Adjust for Add	00100111			
SUB = Subtract:				
Reg./Memory and Register to Either	001010dw	mod reg r/m		
Immediate from Register/Memory	100000sw	mod101r/m	data	data if s w = 01
Immediate from Accumulator	0010110w	data	data if w = 1	
SSB = Subtract with Borrow				
Reg./Memory and Register to Either	000110dw	mod reg r/m		
Immediate from Register/Memory	100000sw	mod011r/m	data	data if s w = 01
Immediate from Accumulator	0001111w	data	data if w = 1	
DEC = Decrement:				
Register/memory	1111111w	mod001r/m		
Register	01001 reg			
NEG = Change sign	1111011w	mod011 r/m		
CMP = Compare:				
Register/Memory and Register	001110dw	mod reg r/m		
Immediate with Register/Memory	100000sw	mod111r/m	data	data if s w = 01
Immediate with Accumulator	0011110w	data	data if w = 1	
AAS = ASCII Adjust for Subtract	00111111			
DAS = Decimal Adjust for Subtract	00101111			
MUL = Multiply (Unsigned)	1111011w	mod100r/m		
IMUL = Integer Multiply (Signed)	1111011w	mod101r/m		
AAM = ASCII Adjust for Multiply	11010100	00001010		
DIV = Divide (Unsigned)	1111011w	mod110r/m		
IDIV = Integer Divide (Signed)	1111011w	mod111r/m		
AAD = ASCII Adjust for Divide	11010101	00001010		
CBW = Convert Byte to Word	10011000			
CWD = Convert Word to Double Word	10011001			

STRING MANIPULATION			
REP = Repeat	1 1 1 1 0 0 1 z		
MOVS = Move Byte/Word	1 0 1 0 0 1 0 w		
CMPS = Compare Byte/Word	1 0 1 0 0 1 1 w		
SCAS = Scan Byte/Word	1 0 1 0 1 1 1 w		
LODS = Load Byte/Wd to AL/AX	1 0 1 0 1 1 0 w		
STOS = Stor Byte/Wd from AL/A	1 0 1 0 1 0 1 w		
CONTROL TRANSFER			
CALL = **Call:**			
Direct within Segment	1 1 1 0 1 0 0 0	disp-low	disp-high
Indirect within Segment	1 1 1 1 1 1 1 1	mod 0 1 0 r/m	
Direct Intersegment	1 0 0 1 1 0 1 0	offset-low	offset-high
		seg-low	seg-high
Indirect Intersegment	1 1 1 1 1 1 1 1	mod 0 1 1 r/m	
JMP = **Unconditional Jump:**	**7 6 5 4 3 2 1 0**	**7 6 5 4 3 2 1 0**	**7 6 5 4 3 2 1 0**
Direct within Segment	1 1 1 0 1 0 0 1	disp-low	disp-high
Direct within Segment-Short	1 1 1 0 1 0 1 1	disp	
Indirect within Segment	1 1 1 1 1 1 1 1	mod 1 0 0 r/m	
Direct Intersegment	1 1 1 0 1 0 1 0	offset-low	offset-high
		seg-low	seg-high
Indirect Intersegment	1 1 1 1 1 1 1 1	mod 1 0 1 r/m	
RET = **Return from CALL:**			
Within Segment	1 1 0 0 0 0 1 1		
Within Seg Adding Immed to SP	1 1 0 0 0 0 1 0	data-low	data-high
Intersegment	1 1 0 0 1 0 1 1		
Intersegment Adding Immediate to SP	1 1 0 0 1 0 1 0	data-low	data-high
JE/JZ = Jump on Equal/Zero	0 1 1 1 0 1 0 0	disp	
JL/JNGE = Jump on Less/Not Greater or Equal	0 1 1 1 1 1 0 0	disp	
JLE/JNG = Jump on Less or Equal/ Not Greater	0 1 1 1 1 1 1 0	disp	
JB/JNAE = Jump on Below/Not Above or Equal	0 1 1 1 0 0 1 0	disp	
JBE/JNA = Jump on Below or Equal/ Not Above	0 1 1 1 0 1 1 0	disp	
JP/JPE = Jump on Parity/Parity Even	0 1 1 1 1 0 1 0	disp	
JO = Jump on Overflow	0 1 1 1 0 0 0 0	disp	
JS = Jump on Sign	0 1 1 1 1 0 0 0	disp	
JNE/JNZ = Jump on Not Equal/Not Zero	0 1 1 1 0 1 0 1	disp	
JNL/JGE = Jump on Not Less/Greater or Equal	0 1 1 1 1 1 0 1	disp	
JNLE/JG = Jump on Not Less or Equal/ Greater	0 1 1 1 1 1 1 1	disp	
JNB/JAE = Jump on Not Below/Above or Equal	0 1 1 1 0 0 1 1	disp	
JNBE/JA = Jump on Not Below or Equal/Above	0 1 1 1 0 1 1 1	disp	
JNP/JPO = Jump on Not Par/Par Odd	0 1 1 1 1 0 1 1	disp	
JNO = Jump on Not Overflow	0 1 1 1 0 0 0 1	disp	
JNS = Jump on Not Sign	0 1 1 1 1 0 0 1	disp	
LOOP = Loop CX Times	1 1 1 0 0 0 1 0	disp	
LOOPZ/LOOPE = Loop While Zero/Equal	1 1 1 0 0 0 0 1	disp	
LOOPNZ/LOOPNE = Loop While Not Zero/Equal	1 1 1 0 0 0 0 0	disp	
JCXZ = Jump on CX Zero	1 1 1 0 0 0 1 1	disp	
INT = **Interrupt**			
Type Specified	1 1 0 0 1 1 0 1	type	
Type 3	1 1 0 0 1 1 0 0		
INTO = Interrupt on Overflow	1 1 0 0 1 1 1 0		
IRET = Interrupt Return	1 1 0 0 1 1 1 1		

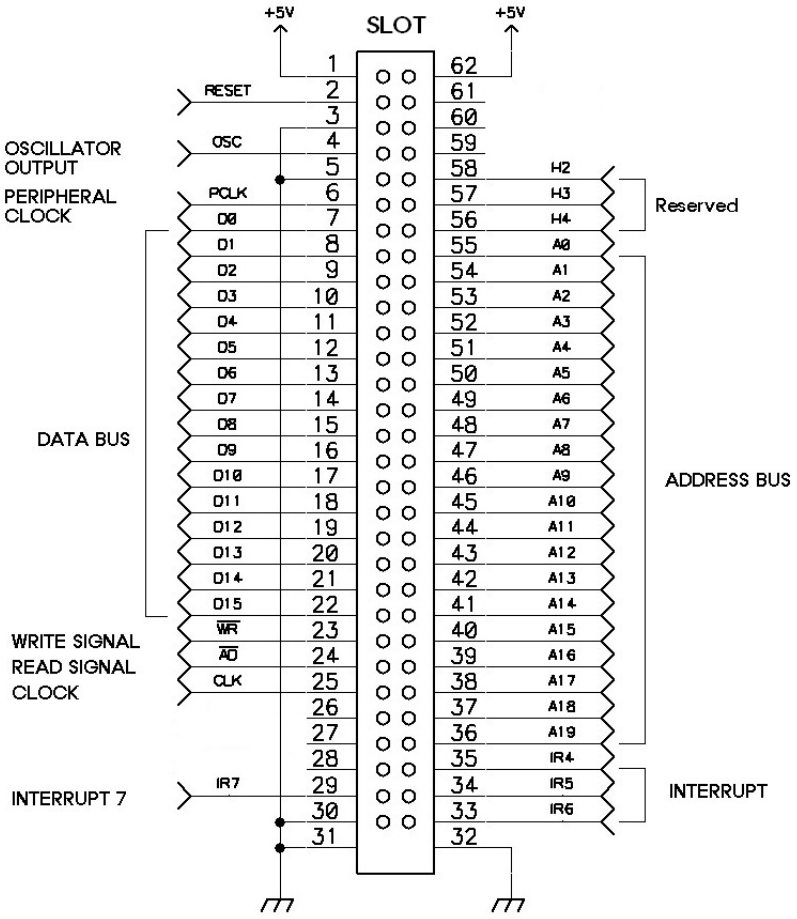
	76543210	76543210
PROCESSOR CONTROL		
CLC = Clear Carry	11111000	
CMC = Complement Carry	11110101	
STC = Set Carry	11111001	
CLD = Clear Direction	11111100	
STD = Set Direction	11111101	
CLI = Clear Interrupt	11111010	
STI = Set Interrupt	11111011	
HLT = Halt	11110100	
WAIT = Wait	10011011	
ESC = Escape (to External Device)	11011xxx	mod x x x r/m
LOCK = Bus Lock Prefix	11110000	

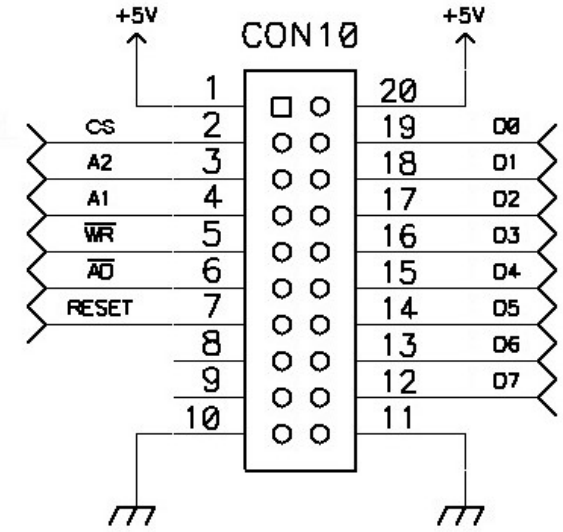
NOTES:
AL = 8-bit accumulator
AX = 16-bit accumulator
CX = Count register
DS = Data segment
ES = Extra segment
Above/below refers to unsigned value
Greater = more positive;
Less = less positive (more negative) signed values
if d = 1 then "to" reg; if d = 0 then "from" reg
if w = 1 then word instruction; if w = 0 then byte instruction
if s w = 01 then 16 bits of immediate data form the operand
if s w = 11 then an immediate data byte is sign extended to form the 16-bit operand
if v = 0 then "count" = 1; if v = 1 then "count" in (CL)
x = don't care
z is used for string primitives for comparison with ZF FLAG

MDA - 8086 Address MAP

Memory address Map		
00000H~0FFFFH	RAM	
10000H~FFFFFH	USER AREA	
F0000H~FFFFFH	ROM	MONITOR PROGRAM
I/O Address Map		
00H 02H 03H 04H	LCD	Even address
01H	Key	Odd address
08H, 0AH	8251	Serial Communication
09H 0BH 0DH, 0FH	8253	Programerable Timer
10H, 12H	8259	Interrupt Controlor
11H	Speaker	
18H 1AH 1CH 1EH	8255	IO
19H 1BH 1DH 1FH	8255	DOT LED

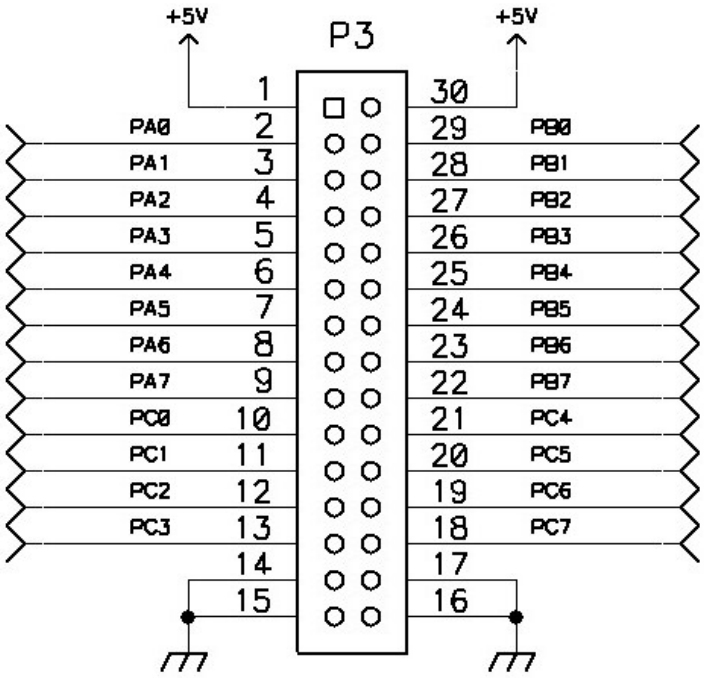
EXT CONNECTOR





Extern 8255 BUS

DO -- D7 : Data Bus from 8086
RESET : Reset Active 'H'
RD : Read Active 'L'
WR : Write Active 'L'
A1 : Address Bus
A2 : Address Bus
CS : Chip Select (20H, 22H, 24H, 26H)



8255 Connect

PA0 -- PA7 : 8255 Port A
PB0 -- PA7 : 8255 Port B
PC0 -- PC7 : 8255 Port C
* Reference : 8255.PDF